CSE470: Software Engineering

Software Requirements Specification (SRS)

End-of-Semester Submission Template

[Fill in every field below. Replace all bracketed instructions with your own content before submission. Delete this instructions line and all other bracketed instruction lines before you submit.]

Project Title

Prepared By

Student ID	Name

Project Information

Section	
Section Faculty	
GitHub Repository Link	
Tech Stack Used	

1. Introduction

1.1 Purpose

[State the purpose of this document and the purpose of the software application it describes. One short paragraph is enough.]

1.2 Scope

[Describe what the application does, who it is for, and what is explicitly out of scope. Include the application name.]

2. Overall Description

2.1 Product Perspective

[Briefly describe how this application fits into a real-world context (e.g. a standalone product, a module of a larger system). One short paragraph.]

2.2 Product High-Level Summary

[List the major requirements of the application in one or two sentences each (these will be broken down into features in Section 4).]

2.3 Users and Characteristics

[List each type of user/actor of the system (e.g. Customer, Admin) and briefly describe their role.]

2.4 Operating Environment

[Describe the platform(s) the application runs on (web browser, Android version, iOS version, desktop OS) and any environment requirements.]

2.5 Design and Implementation Constraints

This project follows the constraints defined in the CSE470 Project Outline, including mandatory MVC architecture and the approved tech-stack policy. State your specific choices below.

[List: (a) the primary tech stack used, (b) the MVC framework/library used and how MVC is structured in your project (Model / View / Controller folders or modules), (c) any AI-backed feature and the FastAPI/AI service used for it, if applicable, (d) any faculty-approved exception.]

2.6 Assumptions and Dependencies (Optional)

[List any assumptions made during development (e.g. availability of a third-party API) and external dependencies.]

3. System Architecture

3.1 Architectural Overview

[Insert or describe an MVC architecture diagram showing how the Model, View, and Controller layers are organized in your project, and how they communicate.]

3.2 Technology Stack

[List each layer of your stack (frontend, backend, database, hosting/deployment) and the specific technology used for each.]

3.3 Class Diagram

[Draw the Class Diagram of your project showing all the classes and relationships necessary to implement your system]

4. Requirements and Features

List every requirement and its associated feature(s) below. Per the Project Outline (Section 3.2.1), all CRUD operations on the same entity by the same actor using the same logic count as ONE feature — do not split them. A minimum of 20 features is required.

[Replace the sample row with your own rows. Add as many rows as needed. Each Feature ID should be unique (F1, F2, F3…) and referenced consistently across this document and your Git commit history.]

Req. ID	Requirement	Feature Description	Actor	MVC Components Touched	Sprint Completed	Status
R1	Example: Customers can manage products in their cart	Example: Add, view, update quantity, and remove products in the cart (single CRUD feature)	Customer	Model: Cart, CartItem; View: CartPage; Controller: CartControll er	Sprint 2	Done
R…	…	…	…	…	…	…
R…	…	…	…	…	…	…

4.1 Feature Count Summary

Minimum features required (per Project Outline, Section 7)	
Total features listed in Section 4 above	
Total features fully completed (“Done” per Section 7.2 of the Project Outline)	

5. External Interface Requirements (Optional)

Interface Type	Description
User Interface	
Hardware Interface	

Interface Type	Description
Software Interface (APIs, third-party services)	
Communication Interface	

6. Non-Functional Requirements (Optional)

Category	Requirement
Performance	
Security	
Usability	
Reliability / Availability	
Scalability	

7. Sprint History and Review Ceremony Log

Record the goals and attainment shared during each weekly sprint review ceremony, per Section 7.1 of the Project Outline.

Sprint	Goals Set (Start of Week)	Attainment Reported (End of Week)	Features Completed This Sprint
Sprint 1			
Sprint 2			
Sprint 3			
Sprint 4			

7.1 Deviations and Lessons Learned

[Briefly explain any sprint where attainment did not match the planned goals, why it happened, and what was adjusted for the following sprint.]

8. Testing Summary

[Briefly describe how each major feature was tested (manual testing steps, sample test cases, or automated tests). A short table or list is acceptable.]

9. AI Tool Usage Disclosure

Per the Project Outline's Academic Integrity & AI Usage Policy (Section 6), disclose every instance where a generative AI tool was used during development. Leave the table empty (do not delete it) if no AI tool was used.

Feature / Task	AI Tool Used	How It Was Used (be specific)	I Can Fully Explain This Code (Yes/No)

[By submitting this document, you confirm that you are able to personally explain, modify, and debug every piece of code listed above during your viva.]

10. Individual Contribution Summary

[Since each student develops their project individually, summarize your own work: which features you implemented, approximate time spent per sprint, and any challenges you solved independently. If you collaborated with your 4-student group on non-code matters (e.g. idea discussion), state that clearly and distinguish it from your individual implementation work.]

11. Pre-Submission Checklist

Checklist Item	Confirmed (Yes/No)
The application implements the MVC architecture	
At least 20 features are listed and each is correctly counted per the granularity rule	
No forbidden framework (e.g. Django, Flask as primary backend) was used	
All four sprints' goals and attainment are recorded in Section 7	
The GitHub repository is up to date and the section faculty is added as a collaborator	
The AI Usage Disclosure table (Section 10) is complete and honest	
I can personally explain and defend every feature listed in this document during the viva	